SCHOOL MANAGEMENT SYSTEM FOR KYAMATU PRIMARY SCHOOL

PROJECT PROPOSAL

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Registration Number: 23/05037

Submission Date: April 2026

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A project document submitted in partial fulfillment of the requirements for the award of the degree of Bachelor of Information Technology (BIT 3105)

DECLARATION

This project proposal is my original work and has not been presented for a degree in any other University.

Signature: _____ Date: _____

Mwange Musa Muthami (23/05037)

Approval:

This proposal has been submitted for examination with my approval as the University supervisor.

Signature: _____ Date: _____

Dr. Simon N. Mwendia / Dr. Kevin Mugoye Sindu

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CHAPTER ONE: INTRODUCTION

1.1 Background

Kyamatu Primary School, situated in Voo, Kitui County, provides foundational education under Kenya's Competency-Based Curriculum (CBC) framework. The institution's daily operations—ranging from student enrollment and curriculum tracking to financial management—are highly data-intensive. Currently, the administration relies exclusively on conventional paper-based registries. Student admission records, attendance sheets, and academic grading matrices are curated in physical files, while the bursar's office manages school levies via manual ledgers and receipt books. This traditional paradigm significantly impacts the speed and transparency of administrative workflows, limiting the school's ability to provide timely academic and financial feedback to parents.

1.2 Problem Statement

The continued use of manual, paper-based administrative systems at Kyamatu Primary School creates severe operational bottlenecks and data vulnerabilities. Physical records are highly susceptible to loss, damage, and illicit alteration. Administrators and teaching staff spend an estimated 30% of their working hours manually transcribing attendance sheets, computing CBC competency grades (Exceeding, Meeting, Approaching, or Below Expectations), and reconciling fee balances.

Furthermore, this manual ecosystem isolates parents from real-time student profiling. Parents currently have no mechanism to remotely verify their child's daily attendance, academic progress, or fee arrears, requiring physical school visits. According to recent educational ICT frameworks, reliance on fragmented physical records results in a 15% rate of data inconsistency across term transitions (Ministry of Education, 2023).

1.3 Proposed System

The proposed solution is a robust, modular, web-based School Management System tailored explicitly for Kyamatu Primary School. This cloud-hosted platform digitizes and integrates core school workflows:

- Role-Based Portals: Secure dashboards for Administrators, Teachers, Bursars, Students, and Parents.
- CBC Assessment Engine: Automated grading algorithm evaluating formative scores against CBC competency parameters.
- Attendance Tracking: Digital attendance recording with real-time statistics and reporting.
- Financial Management: Fee structure configuration, invoice generation, and payment tracking with M-Pesa readiness.

1.4 Objectives

System Objectives

- To develop a centralized cloud architecture that unifies student profiles, staff allocations, and academic records.
- To automate the computation of terminal grades into compliant CBC report cards, reducing compilation time by over 80%.
- To implement a digital attendance tracking module that records and aggregates daily student presence.
- To integrate a digital payment tracking system to process and reconcile school fee settlements.

Research Objectives

- To investigate data loss points and administrative delays in current manual systems at Kyamatu Primary School.
- To evaluate digital literacy and hardware accessibility of teaching staff and parental demographic.
- To design a scalable Entity-Relationship framework that accurately models CBC academic tracking.
- To implement and test a full-stack web application utilizing React.js, Express, and PostgreSQL.

1.5 Significance of the Project

To the Organization: The platform guarantees data integrity, significantly curbing man-hours lost to manual computation and ledger balancing. It equips administrators with analytical dashboards for swift, evidence-based decisions.

To the Society (Parents/Students): It establishes a communication bridge, empowering parents with remote visibility into their children's academic milestones, attendance, and fee status.

CHAPTER TWO: LITERATURE REVIEW

The necessity for localized, cloud-based School Management Systems is paramount for developing rural educational infrastructure. Two prominent systems were reviewed:

2.1 ShulePro

ShulePro is a legacy desktop-based school administration software widely utilized in East Africa for exam processing and fee tracking (ShulePro, 2021). As a predominantly localized desktop application, it lacks cloud-native architecture and remote parental access portals. Furthermore, it operates on standard summative grading metrics, lacking automated CBC formative assessment tracking.

2.2 Zeraki Analytics

Zeraki Analytics is an advanced, cloud-based academic tracking service primarily serving secondary schools across Kenya. Its complex interface is optimized for extensive secondary school ranking structures, making it overly intricate for primary school infrastructure. Critically, it lacks an integrated, user-facing financial management portal for direct fee settlements.

CHAPTER THREE: METHODOLOGY

3.1 Research Methodology

Data Collection: Qualitative semi-structured interviews and ethnographic observation with the Headteacher, Bursar, and homeroom teachers.

Target Population: Administrative and teaching corpus of Kyamatu Primary School, alongside a representative subset of the guardian body.

Sampling: Non-probability purposive sampling to select key informants who process academic and financial data.

3.2 Development Methodology

The software engineering phase is governed by the Agile Development Methodology (Scrum framework). Development is divided into functional Sprints:

- Sprint 1: User Authentication and Database Architecting (Prisma/PostgreSQL)
- Sprint 2: Student Registration, Attendance, and CBC Grading engines
- Sprint 3: Financial Management, Report Generation, and Timetabling

3.3 Budget and Resources

Item	Detail	Cost (KES)
Development Workstation	8GB RAM, i5, SSD	65,000
Testing Device	Android Smartphone	20,000
Frontend Hosting	Cloudflare Pages (Free Tier)	0
Backend Hosting + DB	Render + Supabase	6,000
Internet Access	Monthly ISP	5,000
Printing & Binding	Documents	3,000
Total		99,000

3.4 Project Schedule

Task	Hours	Start	End	Deliverable

Proposal	40	01/09/2025	15/09/2025	Proposal Document
SRS	30	16/09/2025	30/09/2025	SRS Document
Design	45	01/10/2025	20/10/2025	Design Docs
Test Plan	15	21/10/2025	30/10/2025	Test Plan
Implementation	120	01/11/2025	30/01/2026	Source Code
Maintenance Plan	10	01/02/2026	05/02/2026	Maintenance Schedule
User Manual	20	06/02/2026	15/02/2026	User Manual
Final Report	30	16/02/2026	28/02/2026	Final Report

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